

Challenge (Habanero)

- Q1.** What is the absolute value of -5 ?
(A) -5
(B) 5
(C) 0
(D) 10
(E) 15
- Q2.** In a game, a player's score can be represented as x . If $|x| = 10$, what are the possible values of x ?
(A) $0, 10$
(B) $-10, 10$
(C) $5, 15$
(D) $20, 30$
(E) $-5, 5$
- Q3.** A coding error causes a program to use $|-3|$ times more data than intended. How much more data is used if the intended amount is 9?
(A) 3
(B) 6
(C) 9
(D) 12
(E) 15
- Q4.** The absolute value of x is 4. What is $|x + 2|$ if $x = -6$?
(A) 2
(B) 4
(C) 6
(D) 8
(E) 10
- Q5.** If $|x - 5| = 2$, what are the possible values of x ?
(A) $3, 7$
(B) $5, 10$
(C) $2, 4$
(D) $1, 3$
(E) $6, 8$
- Q6.** A gamer's score is represented by $|2x + 1|$. If the score is 11, what is x ?
(A) 5
(B) -5
(C) 4
(D) -4
(E) 3
- Q7.** If $|x| = |2x - 3|$, what is x ?
(A) 1
(B) -1
(C) 2
(D) 0
(E) 3
- Q8.** The absolute value of the difference between two gamers' scores is 8. If one score is 12, what is the other score?
(A) 20
(B) 4
(C) 16
(D) 8
(E) 24

Open Ended Questions (FRQ)

- Q9.** A programmer writes a code to calculate the absolute value of x . If x represents the number of points a player is ahead or behind in a game, and x can be any integer between -10 and 10 , describe how the absolute value function $|x|$ relates to the player's position in the game. Provide at least 3 examples.
- Q10.** Design a simple game where the player's score is determined by the absolute value of x . The game ends when the score reaches 20. Describe the game rules, including how x is determined and how the absolute value affects the gameplay. Include a scoreboard example.

Chef's Tips

- Ensure students understand that absolute value represents distance from zero on the number line.
- Emphasize the importance of considering both positive and negative values when evaluating absolute value expressions.
- Remind students to read each question carefully and choose the correct answer based on the given context.